

An update to ECMWF's machine-learned weather forecast model AIFS

Source Information

"An update to ECMWF's machine-learned weather forecast model AIFS", Gabriel Moldovan, Ewan Pinnington, Ana Prieto Nemesio et al., ECMWF, arXiv:2509.18994v1, March 2025. Preprint, 22 pages. Describes AIFS 1.1.0, released 27 August 2025.

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update_to_ECMWF_machine_learned_weather_forecast_model_aifs Original: 2509.18994v1, p.1

Research Objective

To improve the operational deterministic AIFS Single along three axes at once — physical consistency, training schedule, and variable coverage — without regressing the headline scores. The paper frames the motivation partly around cost: MSE-trained models are much cheaper to train than probabilistically trained ones, which makes them attractive for prototyping (2509.18994v1, p.2).

Problem

Machine-learned forecast models produce outputs that violate known physical relationships and limits — negative precipitation, mass imbalances. Until now this was handled by post-processing forecasts to remove the inconsistencies (2509.18994v1, p.2). Separately, such models cover only a limited subset of available forecast variables, which limits their usefulness for downstream sectors.

Gap Addressed in paper

Two things, and the first is the real contribution. Rather than post-process unphysical output, the model gets a final layer of activation functions that bound variables within physically meaningful limits and enforce constraints between related quantities. This constrains the output space to physically plausible regimes before training rather than repairing it after (2509.18994v1, p.1).

The second is variable coverage. For the first time in AIFS: soil moisture, soil temperature and runoff, plus energy-sector variables including cloud cover, 100 metre winds and solar radiation, and snowfall (2509.18994v1, p.2).

Findings and conclusion

Improvements of around 4-6% across all variables, lead times and pressure levels relative to the previous version. The largest gains — up to 12% in normalised difference in the short range — are in total precipitation, which the authors attribute to the new bounding (2509.18994v1, p.16).

The stated mechanism for the precipitation gain is worth noting because it is a claim about representation, not about accuracy. The bounding layer maps negative outputs to no-rain, so the negative space acts as learned likelihood of no-rain and the model no longer has to learn to output exactly zero. The authors hypothesise this is why no-rain and light precipitation improved specifically, and support it with an ablation showing the negative-space behaviour once the final bounding layer is removed (2509.18994v1, p.15).

Upper-air headline scores also improved, and the authors state that most non-precipitation gains stem from the expanded training dataset — ERA5 extended from 1979-2020 to 1979-2022 — and from using recent operational analyses for rollout fine-tuning, rather than from any architectural change (2509.18994v1, p.16).

The training schedule also simplified: operational analysis is now used for the whole fine-tuning stage instead of ERA5 first then operational, which the authors say reduces cost and improves performance (2509.18994v1, p.3).

Limitations or Weakness

The most interesting admission is that rollout fine-tuning *worsens* smoothing. Blurring is already present after pre-training, and rollout fine-tuning enhances it, because minimising MSE at lead times up to 72 h inevitably blurs. The authors are explicit that learning-rate scheduling, number

of steps and rollout strategy all affect blur intensity, and that the effect is not yet understood (2509.18994v1, p.17).

This produces a named trade-off: forecast realism versus optimisation of MSE-based verification scores. Aggressive rollout strategies can significantly boost headline scores; this team deliberately chose a compromise that keeps the spectral signature closer to the analysis, accepting lower RMSE (2509.18994v1, p.18). A reader should note that this is a deliberate choice to report worse headline numbers, which is rare and creditable — but it also means the reported 4-6% is not the ceiling this architecture can reach.

The bounding layer hard-constrains outputs, so information in the negative space cannot propagate into the weights. The authors identify this as a limitation and plan to investigate LeakyReLU-based alternatives that permit weight updates there (2509.18994v1, p.17).

Also open: whether the latent space must grow to accommodate the added earth-system and energy-sector variables, and whether the new variables — currently drawn from the same data source — can eventually be replaced by observation-based data.

Context worth recording: AIFS 1.0.0 became operational on 25 February 2025, and 1.1.0 was released on 27 August 2025 specifically to correct a precipitation forecast issue in the initial version (2509.18994v1, p.1).

Implication or suggestions on future research

LeakyReLU-style bounding that still allows gradient flow in the negative space. Systematic study of which training hyperparameters control field smoothing. Ocean and wave components, expanded cryospheric processes, and increased hydrological capability. The authors stress that regular fine-tuning on up-to-date ECMWF analyses is necessary, because the model rides on those analyses for real-time forecasting and they shift with each IFS model cycle.

How your search can fill gap

The bounding-layer result is the most transferable idea here and the easiest to test outside ECMWF. The claim that a hard output bound improves a non-Gaussian variable by giving the negative space a meaning is a falsifiable mechanism, not just an empirical score. Reproduce it on a third variable with a comparable distribution — convective precipitation is the obvious candidate — and compare a hard ReLU bound, a LeakyReLU bound and no bound under an

identical budget. If the mechanism is real, the hard bound should win specifically on the no-rain/light-rain discrimination and the gap should narrow when LeakyReLU is given gradient access.

Second, the realism-versus-RMSE trade-off is stated but not quantified as a curve. Sweeping rollout length against spectral realism would turn a stated trade-off into a measured one, and would answer the first AIFS paper's open question about how optimisation window length controls smoothing.